

Quantum SVM Application Crash Report

Date: August 9, 2025

Application: svm_quantum/main.py

Environment:

- **Operating System:** Windows
- **Python Version:** 3.13 (inferred from traceback path)
- **Libraries:** qiskit, qiskit-machine-learning, numpy

1. Overview

The application `svm_quantum/main.py` terminated unexpectedly during the model training phase. The crash occurred while the program was attempting to compute the kernel matrix for a Quantum Support Vector Machine (QSVM). The core issue stems from an underlying memory allocation failure within the `numpy` library.

2. Incident Details

The program execution was halted by a `numpy._core._exceptions._ArrayMemoryError`. This exception was the direct cause of a higher-level `qiskit_machine_learning.exceptions.AlgorithmError: 'Sampler job failed!'`.

Full Traceback:

Traceback (most recent call last):

File "D:\MajorProject\test\Disease_classifier_classical\SVM@\comparision\svm_quantum\main.py", line 56, in <module>

main()

File "D:\MajorProject\test\Disease_classifier_classical\SVM@\comparision\svm_quantum\main.py", line 41, in main

model.fit(X_train, y_train)

File "C:\Users\Hp\AppData\Roaming\Python\Python313\site-packages\sklearn\base.py", line 1389, in wrapper

return fit_method(estimator, *args, **kwargs)

File "C:\Users\Hp\AppData\Roaming\Python\Python313\site-packages\sklearn\svm_base.py", line 258, in fit

fit(X, y, sample_weight, solver_type, kernel, random_seed=seed)

... (intermediate calls) ...

File

"C:\Users\Hp\AppData\Roaming\Python\Python313\site-packages\qiskit\quantum_info\states\quantum_state.py", line 371, in _index_to_ket_array

str_kets = np.char.add(row, str_kets)

`numpy._core._exceptions._ArrayMemoryError: Unable to allocate 3.94 MiB for an array with shape`

(4096,) and data type <U252

The above exception was the direct cause of the following exception:

Traceback (most recent call last):

File "D:\MajorProject\test\Disease_classifier_classical\SVM@\comparision\svm_quantum\main.py", line 56, in <module>

main()

... (intermediate calls) ...

qiskit_machine_learning.exceptions.AlgorithmError: 'Sampler job failed!'

3. Root Cause Analysis

The core problem is a **memory allocation failure**. The program ran out of available memory while trying to create a `numpy` array. This happened deep within the Qiskit library, specifically in the `qiskit.quantum_info.states.quantum_state.py` module.

The traceback indicates the code was attempting to create a NumPy array with:

- **Shape:** (4096,)
- **Data Type:** <U252 (a Unicode string of up to 252 characters)
- **Size:** 3.94 MiB

The shape (4096,) is a significant clue. In quantum computing, $2^{12}=4096$, which strongly suggests that the quantum circuit being used has **12 qubits**. As the number of qubits increases, the computational resources required grow exponentially.

The program was trying to generate string representations for all 212 possible quantum states (e.g., '000...00', '000...01', etc.). However, for some reason, the required string size <U252 is unusually large. This could be:

1. **A bug in the specific version of Qiskit** or its dependencies running on Python 3.13, causing an oversized string to be allocated.
2. **An issue with a high-dimensional feature map** that uses a large number of qubits, exceeding the available memory of the system.
3. **An incompatibility** between the installed versions of `numpy` and `qiskit-machine-learning`.

The higher-level error, `AlgorithmError: 'Sampler job failed!'`, is a direct result of this low-level memory failure, as the Sampler primitive was unable to complete its task.

4. Recommendations & Next Steps

To resolve this issue, consider the following actions:

1. **Reduce the problem size.** The most direct solution is to work with a smaller dataset or a feature map that uses fewer qubits. This will drastically reduce the memory required for state representations. For example, using a feature map with 8 qubits ($2^8=256$ states) would be much more manageable.
2. **Upgrade your Python libraries.** The traceback shows several `DeprecationWarning` messages, indicating that the `Sampler` primitive is outdated. It is highly recommended to upgrade your Qiskit libraries to the latest stable versions. This can be done with pip:

```
pip install --upgrade qiskit qiskit-machine-learning numpy
```

Newer versions may contain bug fixes for memory management and will use the more efficient **V2 primitives** (like `StatevectorSampler`) as suggested by the warning.

3. **Use a different primitive.** As noted in the warnings, the `Sampler` class is deprecated. Explicitly switch to the recommended `StatevectorSampler` when initializing your `ComputeUncompute` class.
4. **Check system resources.** Verify that your system has enough available RAM to handle the operations. While 3.94 MiB is not a large amount, it could be a symptom of a larger memory leak or a system with very limited resources. The problem may not be the array size itself, but the fact that it was the final straw that broke the available memory.